

# OligoTox-CNS — Source Register

NIH/NCATS Oligonucleotide Toxicity Open Data Challenge — Phase 2 · CNS / neurotoxicity module  
· companion to the dataset, narrative, methodology and PADP

**What this document is.** Every one of the 4,428 measurements in this dataset was read out of a published source. This register names all of them: the database each file came from, the exact endpoint that returned it, the table or figure inside it that was read, the licence it carries, and how many released rows it accounts for. It also lists what was retrieved and *not* used, and why.

**Nothing here is hand-typed.** The source blocks in §3, the trial table in §4 and every count in this document are computed at build time from the released rows, from the source registry in `src/assemble.py`, and from the retrieved JSON on disk. Regenerate with `python3 src/make_sources.py`.

|                                                          | count |
|----------------------------------------------------------|-------|
| Sources contributing at least one released row           | 8     |
| Sources contributing measurement instruments only        | 1     |
| Oligonucleotides                                         | 1,879 |
| CNS toxicity measurements                                | 4,428 |
| Clinical trial records retrieved                         | 29    |
| of which have posted results, and were read              | 22    |
| Sources retrieved and held, contributing no row (§5)     | 5     |
| Sources examined and rejected on the record (§6)         | 4     |
| Human <i>in vitro</i> sources identified and queued (§7) | 8     |

*Every figure in this table is counted from the released data at build time.*

## 1 Databases and interfaces queried

Nine distinct services were used. The endpoint column gives the address actually called, not a landing page, so any row in this dataset can be traced back to a request a reader can repeat.

| Database                              | What it is                                                                                                                                                                                                           | Endpoint called                                                                                                                                                                                                           | Used for                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Europe PMC REST — full text           | European Bioinformatics Institute (EMBL-EBI). Returns the publisher-deposited JATS XML of an open-access article: body, methods, tables, figure legends and the machine-readable permissions block.                  | <a href="https://www.ebi.ac.uk/europepmc/webservices/rest/&lt;PMCID&gt;/fullTextXML">https://www.ebi.ac.uk/europepmc/webservices/rest/&lt;PMCID&gt;/fullTextXML</a>                                                       | H1, HV1, HV3, and the licence verification for every source in §3                         |
| Europe PMC REST — supplementary files | The same service, returning a ZIP of the article's supplementary material. This is the only programmatic route to supplementary tables, which is where sequence-resolved oligonucleotide data almost always lives.   | <a href="https://www.ebi.ac.uk/europepmc/webservices/rest/&lt;PMCID&gt;/supplementaryFiles">https://www.ebi.ac.uk/europepmc/webservices/rest/&lt;PMCID&gt;/supplementaryFiles</a>                                         | H1 Table S1 (the module's core), K1 Table S2, L1 Table S1, HV2 source data, HV3 mmc1/mmc3 |
| Europe PMC REST — search              | Fielded search over 45+ million records (TITLE:, AUTH:, JOURNAL:, DOI:, PMCID:, PUB_YEAR:). <code>resultType=core</code> returns the DOI, PMID, PMCID, open-access flag and licence in one call.                     | <a href="https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=...&amp;resultType=core&amp;format=json">https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=...&amp;resultType=core&amp;format=json</a> | discovery of every literature source; identifier and licence verification in §8           |
| PubMed Central                        | NIH/NLM's full-text archive. Used to read articles in rendered form and to confirm the copyright statement shown to a human reader.                                                                                  | <a href="https://pmc.ncbi.nlm.nih.gov/articles/&lt;PMCID&gt;/">https://pmc.ncbi.nlm.nih.gov/articles/&lt;PMCID&gt;/</a>                                                                                                   | orientation and cross-checking for all PMC-deposited sources                              |
| ClinicalTrials.gov API v2             | NIH/NLM trial registry. <code>resultsSection.adverseEventsModule</code> carries the full MedDRA adverse-event table for a trial with posted results: every term, every arm, with both a numerator and a denominator. | <a href="https://clinicaltrials.gov/api/v2/studies/&lt;NCT&gt;">https://clinicaltrials.gov/api/v2/studies/&lt;NCT&gt;</a>                                                                                                 | CT1 — the module's principal human source; see §4 for the trial-by-trial table            |

| Database                          | What it is                                                                                                                                                                                 | Endpoint called                                                                   | Used for                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| DailyMed                          | NIH/NLM's archive of current FDA-approved product labelling, addressed by a stable <code>setid</code> . The approved label is the regulatory statement of a drug's adverse-effect profile. | <code>https://dailymed.nlm.nih.gov/dailymed/lookup.cfm?setid=&lt;setid&gt;</code> | C1 — QALSODY (tofersen) and SPINRAZA (nusinersen) prescribing information |
| WHO INN (Recommended List)        | The World Health Organization's International Nonproprietary Name entry, which for an oligonucleotide prints the full sequence and the per-position chemistry as a formal description.     | WHO Drug Information, INN Recommended List (tofersen)                             | the tofersen sequence and its 15 PS / 4 PO linkage pattern                |
| USPTO full-text (via patent PDF)  | Granted US patents. Patent tables routinely pair an oligonucleotide sequence with a toxicity rating, and are US Government works in the public domain.                                     | US 10,799,523 B2 (Olson et al.)                                                   | P1 — retrieved and held, not yet ingested (\$5)                           |
| Publisher sites (OUP, NEJM, CHDI) | Read directly where the article or conference deck is not deposited in PMC.                                                                                                                | <code>academic.oup.com/nar/...</code> ; <code>nejm.org</code>                     | O1 cross-check; M1, S1 and B2 read but not redistributable (\$5)          |

## 2 What each route does and does not return

Six retrieval routes return something that looks like success and is not. Each cost real time to find, and each would have put wrong data — or no data — into the dataset silently. They are recorded because the next person to build on these sources will meet them too.

| Trap                                                                      | What happens                                                                                                                                                                                                                                                                | What this pipeline does                                                                                                                         |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PMC binary downloads are gated</b>                                     | Requesting a supplementary binary directly from <code>pmc.ncbi.nlm.nih.gov</code> returns <b>HTTP 200 with an HTML challenge page</b> , not the file. A pipeline that trusts the status code silently writes a JavaScript proof-of-work page to disk and parses it as data. | Every download is checked with <code>file</code> before it is parsed. Supplementary material is taken from the Europe PMC endpoint instead.     |
| <b>The ClinicalTrials.gov results page has no results in it</b>           | <code>clinicaltrials.gov/study/&lt;NCT&gt;?tab=results</code> is a client-side application. Its HTML contains none of the adverse-event text, so scraping the page returns nothing while appearing to succeed.                                                              | All CT1 data is taken from API v2 and the returned JSON is committed to the repository, so the build does not depend on the service staying up. |
| <b>The Europe PMC supplementary ZIP can carry another article's files</b> | A first call for HV1 returned a valid 898 kB ZIP, HTTP 200 — whose 47 members all belonged to a different paper. This is a cache collision, and it is invisible unless the member filenames are inspected.                                                                  | Every extracted ZIP is checked member-by-member against the expected article prefix before anything inside it is read.                          |
| <b>Plain text extraction destroys chemistry</b>                           | Source L1 encodes each nucleotide's chemistry in the <b>typeface</b> of its supplementary table — bold = LNA, bold-italic = 2'-MOE, regular = DNA. Copying the text out yields a sequence with the chemistry silently deleted.                                              | The parser reads PDF span styling, so per-position chemistry is recovered mechanically rather than transcribed by eye.                          |
| <b>Sequences that exist only in the supplement</b>                        | All 23 of source HV3's modified sequences appear in <code>mmc1.pdf</code> and <code>mmc3.pdf</code> and <b>nowhere in the article text</b> . A full-text-only extraction finds none of them and concludes the paper prints no sequences.                                    | Supplementary material is retrieved for every source, not only where the text implies it exists.                                                |
| <b>The host matters</b>                                                   | <code>www.ncbi.nlm.nih.gov/pmc/articles/&lt;PMCID&gt;/</code> now answers with a 301 redirect; the working host is <code>pmc.ncbi.nlm.nih.gov</code> .                                                                                                                      | Recorded here so a future run does not read the redirect as an absent article.                                                                  |

## 3 The sources, one by one

Each block below is generated from the source registry and the released rows. **Contributes** is counted from the data, so a source cannot appear here claiming rows it did not supply.

## H1 Hagedorn 2022

|               |                                                                                                                                                                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | Hagedorn PH, Brown JM, Easton A, Pierdomenico M, Jones K, Olson RE, Mercer SE, Li D, Loy J, Hog AM, Jensen ML, Gill M, Cacace AM. Acute Neurotoxicity of Antisense Oligonucleotides After Intracerebroventricular Injection Into Mouse Brain Can Be Predicted from Sequence Features. Nucleic Acid Ther. 2022 Jun;32(3):151-162.              |
| Identifiers   | doi:10.1089/nat.2021.0071 · PMID 35166597 · PMC9221153                                                                                                                                                                                                                                                                                        |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC9221153/">https://pmc.ncbi.nlm.nih.gov/articles/PMC9221153/</a>                                                                                                                                                                                                                             |
| Retrieved via | Europe PMC supplementaryFiles REST endpoint                                                                                                                                                                                                                                                                                                   |
| Licence       | <b>CC BY 4.0</b> → cc_by                                                                                                                                                                                                                                                                                                                      |
| Read from it  | Supplementary Table S1 — 1,825 oligonucleotides, each with its full sequence (upper case encoding LNA position, lower case DNA), a measured rat-cortical-neuron calcium-oscillation score, and for 181 of them a measured mouse acute tolerability score. Grade cut-offs 4 / 7 / 18 are the authors' own, quoted from the Fig. 1B discussion. |
| Contributes   | 1,825 oligonucleotides · 2,006 measurements (animal invitro 1,825, animal invivo 181)                                                                                                                                                                                                                                                         |

## K1 Miller 2024

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | Miller BR, Paquette JD, Barker AR, et al. Preventing acute neurotoxicity of CNS therapeutic oligonucleotides with the addition of Ca <sup>2+</sup> and Mg <sup>2+</sup> in the formulation. Mol Ther Nucleic Acids. 2024;35(4):102359. Journal record PMC11567125; the supplementary file read here was retrieved from the bioRxiv preprint record PMC11185713 (doi 10.1101/2024.06.06.597639), which carries the same title, the same authors and the same Table S2. |
| Identifiers   | doi:10.1016/j.omtn.2024.102359 · PMID 39554992 · PMC11567125                                                                                                                                                                                                                                                                                                                                                                                                          |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC11567125/">https://pmc.ncbi.nlm.nih.gov/articles/PMC11567125/</a>                                                                                                                                                                                                                                                                                                                                                   |
| Retrieved via | Europe PMC supplementaryFiles REST endpoint on PMC11185713 (media-1.pdf)                                                                                                                                                                                                                                                                                                                                                                                              |
| Licence       | <b>CC BY-NC-ND 4.0</b> → summary_stat_only                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Read from it  | Supplementary Table S2 (media-1.pdf) — per-group average acute tolerability score, dose in nmol and µg, injectate Ca <sup>2+</sup> and Mg <sup>2+</sup> concentration, group size, mouse genotype, and the paper figure each group appears in.                                                                                                                                                                                                                        |
| Contributes   | 7 oligonucleotides · 41 measurements (animal invivo 41)                                                                                                                                                                                                                                                                                                                                                                                                               |

## L1 Kuroda 2025

|               |                                                                                                                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | Kuroda T, Yoshioka K, Lei Mon SS, Katsuyama M, Sato K, Isogai E, Yoshida-Tanaka K, Iwata-Hara R, Yamaguchi T, Obika S, Yokota T. Unraveling and controlling late-onset neurotoxicity of antisense oligonucleotides through strategic chemical modifications. Mol Ther Nucleic Acids. 2025;36. |
| Identifiers   | doi:10.1016/j.omtn.2025.102692 · PMID 41467114 · PMC12744863                                                                                                                                                                                                                                  |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12744863/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12744863/</a>                                                                                                                                                                           |
| Retrieved via | Europe PMC supplementaryFiles REST endpoint                                                                                                                                                                                                                                                   |
| Licence       | <b>CC BY 4.0</b> → cc_by                                                                                                                                                                                                                                                                      |
| Read from it  | Supplementary Table S1 (five ASOs, per-position chemistry recovered from typeface) and Figures 1A–F with the Results text. Behavioural scores are published only as figures, so the numeric readouts are recorded NOT_REPORTED and the authors' stated outcome is carried instead.            |
| Contributes   | 5 oligonucleotides · 6 measurements (animal invivo 6)                                                                                                                                                                                                                                         |

## C1 FDA PI DailyMed

|               |                                                                                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | QALSODY (tofersen) prescribing information, Biogen Inc., DailyMed setid 81356b45-1cb7-4eef-88ea-e44cc18b47c5, published 2024-11-18. SPINRAZA (nusinersen) prescribing information, Biogen Inc., DailyMed setid dd70cd5f-b0fc-4ba4-a5ea-89a34778bd94, published 2026-04-06. |
| Identifiers   | —                                                                                                                                                                                                                                                                          |
| Link          | <a href="https://dailymed.nlm.nih.gov/dailymed/lookup.cfm?setid=81356b45-1cb7-4eef-88ea-e44cc18b47c5">https://dailymed.nlm.nih.gov/dailymed/lookup.cfm?setid=81356b45-1cb7-4eef-88ea-e44cc18b47c5</a>                                                                      |
| Retrieved via | DailyMed, read directly                                                                                                                                                                                                                                                    |
| Licence       | <b>US Government work / public domain</b> → public_domain                                                                                                                                                                                                                  |
| Read from it  | Prescribing information sections 5.1–5.3 (warnings), 6.1 (adverse reactions table) and 6.2 (post-marketing). The tofersen sequence and its linkage pattern come from the published INN description, not from the label.                                                    |
| Contributes   | 2 oligonucleotides · 12 measurements (human clinical 12)                                                                                                                                                                                                                   |

## CT1 ClinicalTrials gov

|               |                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | ClinicalTrials.gov posted results, adverse-event tables for 22 trials of oligonucleotides delivered into the central nervous system (nusinersen, tofersen, tominersen, BIIB080/MAPTRx, BIIB105, WVE-120101, WVE-120102, WVE-003). U.S. National Library of Medicine.                                                                                                                              |
| Identifiers   | —                                                                                                                                                                                                                                                                                                                                                                                                 |
| Link          | <a href="https://clinicaltrials.gov/api/v2/studies/NCT03761849">https://clinicaltrials.gov/api/v2/studies/NCT03761849</a>                                                                                                                                                                                                                                                                         |
| Retrieved via | ClinicalTrials.gov API v2, resultsSection.adverseEventsModule; the web results page is a client-side application whose HTML contains none of the adverse-event text                                                                                                                                                                                                                               |
| Licence       | <b>US Government work / public domain</b> → public_domain                                                                                                                                                                                                                                                                                                                                         |
| Read from it  | resultsSection.adverseEventsModule — both seriousEvents and otherEvents, every arm including comparators. Scope: the whole ‘Nervous system disorders’ organ class plus a curated set of CNS terms MedDRA files elsewhere (post-lumbar-puncture syndrome, CSF findings, meningitis, myelitis, papilloedema, hydrocephalus). Cardiac ‘ventricular’ terms are excluded — those are heart ventricles. |
| Contributes   | 6 oligonucleotides · 2,329 measurements (human clinical 2,329)                                                                                                                                                                                                                                                                                                                                    |

## HV1 Buijsen 2024

|               |                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | Buijsen RAM, Alagoz E, van der Graaf LM, et al. Calcium-Enhanced Medium-Based Delivery of Splice Modulating Antisense Oligonucleotides in 2D and 3D hiPSC-Derived Neuronal Models. Biomedicines. 2024;12(9):1933. |
| Identifiers   | doi:10.3390/biomedicines12091933 · PMID 39335447 · PMC11428300                                                                                                                                                    |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC11428300/">https://pmc.ncbi.nlm.nih.gov/articles/PMC11428300/</a>                                                                                               |
| Retrieved via | Europe PMC fullTextXML; all three sequences verified against Table 1                                                                                                                                              |
| Licence       | <b>CC BY 4.0</b> → cc_by                                                                                                                                                                                          |
| Read from it  | Table 1 (three 2′-MOE PS sequences), Figure 1B and Figure 2B category labels read from the figure axes, Results §3.1–3.4, Methods §2.4 and §2.6, and the Figure 1 and 2 legends for n and significance.           |
| Contributes   | 3 oligonucleotides · 9 measurements (human invitro 9)                                                                                                                                                             |

## HV2 Chen 2024 Nature

|               |                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | Chen X, Birey F, Li MY, et al. Antisense oligonucleotide therapeutic approach for Timothy syndrome. Nature. 2024;628:818-825.                                                                                                                                  |
| Identifiers   | doi:10.1038/s41586-024-07310-6 · PMID 38658687 · PMC11043036                                                                                                                                                                                                   |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC11043036/">https://pmc.ncbi.nlm.nih.gov/articles/PMC11043036/</a>                                                                                                                                            |
| Retrieved via | Europe PMC / PMC full text                                                                                                                                                                                                                                     |
| Licence       | <b>CC BY 4.0</b> → cc_by                                                                                                                                                                                                                                       |
| Read from it  | Extended Data Fig. 6b, 6d and 6f — which print no numbers — resolved against the released Source Data workbook 41586_2024_7310_MOESM13_ESM.xlsx, sheet by sheet and column by column. The article does not print the ASO sequences, so those are NOT_REPORTED. |
| Contributes   | 7 oligonucleotides · 8 measurements (human invitro 8)                                                                                                                                                                                                          |

### HV3 Woffindale 2026

|               |                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | Woffindale C, Galindo Riera N, Wood MJA, Varela MA. Design, validation, and functional impact of oligonucleotides for multigene silencing in Alzheimer disease. Mol Ther Nucleic Acids. 2026. |
| Identifiers   | doi:10.1016/j.omtn.2026.102848 · PMID 41732207 · PMC12925542                                                                                                                                  |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12925542/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12925542/</a>                                                                           |
| Retrieved via | Europe PMC supplementaryFiles (mmc1.pdf and mmc3.pdf); the 23 sequences appear in the supplement only, NOT in the article text                                                                |
| Licence       | <b>CC BY-NC-ND 4.0</b> → summary_stat_only                                                                                                                                                    |
| Read from it  | Table S2 in mmc1 . pdf for the 23 modified sequences, cross-checked against mmc3 . pdf; Figures 4D and 4H from the main text; Figures S7D and S8D from the supplement.                        |
| Contributes   | 24 oligonucleotides · 17 measurements (human invitro 17)                                                                                                                                      |

### O1 ORourke 2026

|               |                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Citation      | O'Rourke JJ, Bravo-Hernandez M, et al. Acute neuronal inhibition response caused by phosphorothioate antisense oligonucleotides following local delivery to the central nervous system. Nucleic Acids Res. 2026;54(3):gkaf1333. Publisher page: <a href="https://academic.oup.com/nar/article/54/3/gkaf1333/8415850">academic.oup.com/nar/article/54/3/gkaf1333/8415850</a> . |
| Identifiers   | doi:10.1093/nar/gkaf1333 · PMID 41494985 · PMC12865454                                                                                                                                                                                                                                                                                                                        |
| Link          | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12865454/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12865454/</a>                                                                                                                                                                                                                                                           |
| Retrieved via | Europe PMC supplementaryFiles REST endpoint                                                                                                                                                                                                                                                                                                                                   |
| Licence       | <b>CC BY-NC</b> → cc_by_nc                                                                                                                                                                                                                                                                                                                                                    |
| Read from it  | The rodent intrathecal, rodent intracerebroventricular and non-human-primate intrathecal acute-inhibition scales, transcribed into docs/SCORING_INSTRUMENTS . md. No per-oligonucleotide rows: the supplement does not pair sequences with scores.                                                                                                                            |
| Contributes   | 0 oligonucleotides · 0 measurements — instruments only, no rows                                                                                                                                                                                                                                                                                                               |

## 4 ClinicalTrials.gov, trial by trial

CT1 is the module's principal human source and its only source of quantitative hydrocephalus rows. 29 registry records of oligonucleotides delivered into the central nervous system were retrieved in full and committed to the repository; **22 carry posted results** and were read, contributing 2,329 measurements.

| NCT         | Drug           | Lead sponsor                 | Status     | Rows |
|-------------|----------------|------------------------------|------------|------|
| NCT01703988 | nusinersen     | Biogen                       | Completed  | 28   |
| NCT01839656 | nusinersen     | Biogen                       | Completed  | 26   |
| NCT02193074 | nusinersen     | Biogen                       | Terminated | 4    |
| NCT02292537 | nusinersen     | Biogen                       | Completed  | 4    |
| NCT02386553 | nusinersen     | Biogen                       | Completed  | 50   |
| NCT02462759 | nusinersen     | Biogen                       | Terminated | 15   |
| NCT02519036 | tominersen     | Ionis Pharmaceuticals, Inc.  | Completed  | 66   |
| NCT02594124 | nusinersen     | Biogen                       | Completed  | 234  |
| NCT02623699 | tofersen       | Biogen                       | Completed  | 312  |
| NCT03070119 | tofersen       | Biogen                       | Completed  | 105  |
| NCT03186989 | BIIB080_MAPTRx | Ionis Pharmaceuticals, Inc.  | Completed  | 242  |
| NCT03225833 | WVE-120101     | Wave Life Sciences USA, Inc. | Terminated | 156  |
| NCT03225846 | WVE-120102     | Wave Life Sciences USA, Inc. | Terminated | 224  |
| NCT03342053 | tominersen     | Hoffmann-La Roche            | Completed  | 48   |
| NCT03761849 | tominersen     | Hoffmann-La Roche            | Completed  | 156  |
| NCT03842969 | tominersen     | Hoffmann-La Roche            | Completed  | 87   |

| NCT         | Drug       | Lead sponsor                 | Status     | Rows |
|-------------|------------|------------------------------|------------|------|
| NCT04000594 | tominersen | Hoffmann-La Roche            | Completed  | 21   |
| NCT04089566 | nusinersen | Biogen                       | Completed  | 96   |
| NCT04494256 | BIIB105    | Biogen                       | Terminated | 224  |
| NCT04617847 | WVE-120101 | Wave Life Sciences USA, Inc. | Terminated | 27   |
| NCT04617860 | WVE-120102 | Wave Life Sciences USA, Inc. | Terminated | 84   |
| NCT05032196 | WVE-003    | Wave Life Sciences USA, Inc. | Completed  | 120  |

Each record resolves at <https://clinicaltrials.gov/study/<NCT>> for a human reader and at <https://clinicaltrials.gov/api/v2/studies/<NCT>> for the data. Rows are counted from the released dataset.

**7 completed or terminated CNS oligonucleotide trials have posted no results at all.** Each was retrieved successfully and each returns `hasResults: false`, so the registry holds no adverse-event table to read.

| NCT         | Drug      | Status     |
|-------------|-----------|------------|
| NCT00190424 | CpG-ODN   | Completed  |
| NCT03626012 | BIIB078   | Completed  |
| NCT03976349 | BIIB094   | Completed  |
| NCT04259281 | GTX-102   | Completed  |
| NCT04428281 | RO7248824 | Completed  |
| NCT04442295 | STK-001   | Completed  |
| NCT04931862 | WVE-004   | Terminated |

This is a finding about the field rather than a gap in the retrieval: the safety data for these programmes is not public anywhere this pipeline can reach.

## 5 Retrieved and held, contributing no released row

These were read during research. None supplies a row, so removing every one of them changes nothing in the released dataset. They are listed because a reader is entitled to know what informed the work as well as what is in it.

| ID | Source                                                                                                                                                                                               | Identifiers                                           | Licence                                   | In repo? | What it would add                                                                                                                                                      |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B1 | Bravo-Hernández M, et al. Transient acute neuronal activation response caused by high concentrations of oligonucleotides in the cerebral spinal fluid. <i>Nucleic Acids Res.</i> 2026;54(3):gkag057. | doi:10.1093/nar/gkag057 · PMID 41633500 · PMC12867516 | CC BY-NC 4.0                              | yes      | Would add the first non-human-primate rows and upgrade the acute-activation instrument in docs/SCORING_INSTRUMENTS.md §4 from a fetched summary to a direct reading.   |
| P1 | Olson RE, et al. US Patent 10,799,523 B2 — oligonucleotide compounds for CNS delivery.                                                                                                               | US 10,799,523 B2                                      | US Government work — public domain        | yes      | Patent tables pair a sequence with a toxicity rating. This is the format that supplied 21 rows to the sibling kidney module, so the extraction path is already proven. |
| S1 | Schobel S. <i>Preliminary results from GENERATION HD1</i> . CHDI Therapeutics Conference, 2021.                                                                                                      | conference presentation                               | © Roche — not licensed for redistribution | no       | Tominersen ventricular-volume and neurofilament-light detail behind the clinical failure.                                                                              |
| B2 | Boak L, McColgan P. <i>Treatment and post-treatment effects of tominersen in GENERATION HD1</i> . CHDI Therapeutics Conference, 2022.                                                                | conference presentation                               | © Roche — not licensed for redistribution | no       | Post-treatment follow-up on the same trial.                                                                                                                            |
| M1 | McColgan P, et al. Tominersen in Adults with Manifest Huntington's Disease. <i>N Engl J Med.</i> 2023;389(23):2203–2205 (Correspondence).                                                            | doi:10.1056/NEJMc2300400 · PMID 38055260              | © NEJM — not licensed for redistribution  | no       | The peer-reviewed statement of the GENERATION HD1 outcome.                                                                                                             |

Three of these are on the working disk but deliberately not in the repository. S1, B2 and M1 are copyrighted publisher material not licensed for redistribution — two Roche conference decks and an NEJM correspondence item. Committing them would republish them, which is a different act from reading them for research. They are listed in .gitignore, and named in full above so anyone can retrieve their own copy. This is the same per-source-terms discipline LICENSE.md applies at row level. This was a judgement call, not an instruction. Say the word and they can be committed.

## 6 Examined and rejected, with the reason

A source register that lists only what was used is a reading list, not a provenance record. These were found, assessed and turned down; the reason is stated so the decision can be argued with.

| Source                                                                                                                                                                                     | Identifiers                                                                               | Why it looked right                                                                                                               | Why it was rejected                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drygin D, Barone S, Bennett CF. Sequence-dependent cytotoxicity of second-generation oligonucleotides. <i>Nucleic Acids Res.</i> 2004;32(21):6585–6594.                                    | doi:10.1093/nar/gkh997 · PMID 15604456 · PMC545465                                        | 43 oligonucleotides with per-compound cytotoxicity — the largest human panel found, and it prints every sequence.                 | <b>Wrong organ.</b> The lines are A549 lung, HepG2 and Hep3B liver. Human, but not CNS. Admitting it would put another organ's toxicity inside a CNS module.                                                                    |
| Yuan NY, et al. Neural organoids incorporating microglia to assess neuroinflammation and toxicities induced by known developmental neurotoxicants. <i>Curr Res Toxicol.</i> 2025;9:100252. | doi:10.1016/j.crtox.2025.100252 · PMID 40799410 · PMC12341589                             | A human iPSC neural organoid platform with microglia — exactly the kind of human in vitro system the Challenge brief prioritises. | <b>No oligonucleotide is tested.</b> The compounds are lead acetate and other industrial developmental neurotoxicants. The platform is relevant; the data is not oligonucleotide data.                                          |
| Means JC, et al. Rapid and scalable personalized ASO screening in patient-derived organoids. <i>Nature.</i> 2025;638:237–243.                                                              | doi:10.1038/s41586-024-08462-1 · PMID 39843740 · PMC11798851                              | The highest-profile personalised-ASO organoid screen in the literature.                                                           | <b>Cardiac organoids</b> , from Duchenne muscular dystrophy patients. Recorded here as an explicit exclusion so it is not re-surfaced as a CNS candidate on a later sweep.                                                      |
| Seven ClinicalTrials.gov records of CNS-delivered oligonucleotides — BIIB078, BIIB094, GTX-102, RO7248824, STK-001, WVE-004, and a CpG-ODN glioblastoma trial.                             | NCT03626012, NCT03976349, NCT04259281, NCT04428281, NCT04442295, NCT04931862, NCT00190424 | All were retrieved in full and are committed to the repository.                                                                   | <b>No posted results.</b> Every one carries hasResults: false despite being completed or terminated, so the registry holds no adverse-event table to read. This is a finding about the field, not a retrieval failure — see §4. |

## 7 Human *in vitro* backlog — identified, not yet extracted

The Challenge brief states a particular interest in “datasets based on in vitro human systems or able to extrapolate data between in vitro human systems and animal data”. This release carries **34 human *in vitro* measurements** from sources HV1–HV3. The sources below were found in the same sweep and are queued for the next revision. **No released row depends on any of them**, so this is a costed backlog rather than an unknown.

| Source                                                                                                                                                                                      | Identifiers         | Human system                              | What it carries                                                                                                              | Seq? |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------|
| Flynn LL, et al. Single stranded fully modified-phosphorothioate oligonucleotides can induce structured nuclear inclusions... <i>Front Genet.</i> 2022;13:791416.                           | PMC9019733 · CC BY  | SH-SY5Y; primary human dermal fibroblasts | 90+ sequences, 18–30 nt, four chemistries. The largest sequence-resolved human panel in the backlog.                         | yes  |
| Ottesen EW, Murzyn WA, Kaas RL, Bertrand KJ, Payne JL, Singh RN. A therapeutic antisense oligonucleotide encompassing 2'-O-methoxyethyl modification... <i>NAR Mol Med.</i> 2026;3:ugag002. | PMC12805893 · CC BY | SH-SY5Y; SMA patient fibroblasts          | An 18-mer with sequence and chemistry identical to nusinersen — a marketed CNS ASO already in this dataset's clinical layer. | yes  |

| Source                                                                                                                                                                                                               | Identifiers                  | Human system                                                 | What it carries                                                                           | Seq?       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------|
| Thirumalai S, Livesey FJ, Patani R, Hung C. APP antisense oligonucleotides are effective in rescuing mitochondrial phenotypes in human iPSC-derived trisomy 21 astrocytes. <i>Alzheimers Dement.</i> 2025;21:e14560. | PMC11775556 · CC BY          | hiPSC-derived astrocytes, control and Down syndrome          | One 20-mer gapmer, PS backbone, five 2'-MOE each side, 5-methyl-deoxycytidine in the gap. | yes        |
| Umek T, et al. Oligonucleotides Targeting DNA Repeats Downregulate <i>Huntingtin</i> Gene Expression in Huntington's Patient-Derived Neural Model. <i>Nucleic Acid Ther.</i> 2021;31(6):443–456.                     | PMC8713517 · CC BY           | HD patient iPSC → NSC and neurons (NINDS repository)         | CAG19, a 19-nt DNA/LNA mixmer with a full PS backbone.                                    | yes        |
| Faravelli I, et al. Targeted antisense oligonucleotide treatment rescues developmental alterations in spinal muscular atrophy organoids. <i>Nat Commun.</i> 2025;16:988.                                             | PMC12847787 · CC BY-NC-ND    | SMA patient hiPSC spinal cord and cerebral organoids         | MO-10-34 morpholino targeting SMN2 ISS-N1, bare and r6-conjugated.                        | yes        |
| Bowles KR, et al. ELAVL4, splicing, and glutamatergic dysfunction precede neuron loss in MAPT mutation cerebral organoids. <i>Cell.</i> 2021;184(17):4547–4563.e17.                                                  | PMC8635409 · not open access | FTD patient iPSC cerebral organoids                          | The single ASO-in-cerebral-organoid study identified by the Lange 2022 review.            | yes        |
| König S, et al. Transferrin-Functionalized Liposomes Enhance MAPT-ASO Transport Across a 3D Blood-Brain Barrier Microvascular Model. <i>Int J Mol Sci.</i> 2025;26(23):11347.                                        | PMC12691827 · CC BY          | Microfluidic 3D BBB chip — human BMEC, astrocytes, pericytes | A Cy3-labelled MAPT-targeting PS ASO; the paper prints the sequence.                      | yes        |
| Selvakumaran J, et al. An Induced Pluripotent Stem Cell-Derived Human Blood-Brain Barrier Model... <i>Biomedicines.</i> 2023;11(10):2700.                                                                            | PMC10604610 · CC BY          | iPSC-derived brain microvascular endothelial cells           | Pip6a-PMO targeting SMN2 exon 7.                                                          | not stated |

Every identifier in this table was resolved against the Europe PMC search API rather than carried over from a reading note.

**One lead is recorded here without a verified citation.** A study reported as Yoshikawa 2025, *J Pharmacol Toxicol Methods* 135:107844 — 27 ASOs run through a rat calcium-oscillation IC<sub>50</sub> assay, then mouse ICV, then a human iPSC multi-electrode array — would be the closest match in the literature to the brief's extrapolation clause. Four separate Europe PMC queries (by page number, by author and year, by title phrase, and by DOI) return nothing. It is therefore listed as **an unverified lead to chase, not a citation**, and no part of this dataset rests on it.

## 8 The licence audit that produced this document

Writing this register meant re-resolving every identifier against Europe PMC and reading each article's own machine-readable permissions block rather than trusting the reading notes. That found two errors in the released dataset, both now fixed and both material.

|                       | What the registry said                                                                                                          | What the source actually says                                                                                                                      | Effect                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K1</b> Miller 2024 | CC BY-NC → cc_by_nc; cited under a title the article does not carry; PMCID from the preprint record paired with the journal DOI | <b>CC BY-NC-ND 4.0</b> — verbatim from the permissions block of <i>both</i> the journal record (PMC11567125) and the preprint record (PMC11185713) | NoDerivatives. Its 41 measurements are now summary_stat_only, not redistributable as dataset content. The dataset had been over-claiming its rights. |
| <b>L1</b> Kuroda 2025 | CC BY-NC → cc_by_nc                                                                                                             | <b>CC BY 4.0</b> — “This is an open access article under the CC BY license”                                                                        | Its 6 measurements are now cc_by. The dataset had been under-claiming, needlessly restricting rows the authors released freely.                      |

LICENSE.md was also found quoting row counts from a release three revisions old. It is now generated from the released rows by src/make\_summary.py, so it cannot drift again. The same applies to this document.



## 9 Checking any of this yourself

| To check                                                   | Command                                                                                                               |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| The licence of any source                                  | <code>curl 'https://www.ebi.ac.uk/europepmc/webservices/rest/&lt;PMCID&gt;/fullTextXML'   grep -A3 permissions</code> |
| That a measurement's source and location are recorded      | <code>cut -d, -f1,2,3 toxicity/*/data/measurements.csv</code>                                                         |
| Which rows came from which source                          | <code>python3 qc/validate_dataset.py --json</code>                                                                    |
| An adverse-event table, at source                          | <code>curl 'https://clinicaltrials.gov/api/v2/studies/&lt;NCT&gt;'   jq .resultsSection.adverseEventsModule</code>    |
| That the whole dataset rebuilds from the committed sources | see README.md 'Rebuilding from scratch' – eight commands, no network needed                                           |
| This document                                              | <code>python3 src/make_sources.py</code>                                                                              |

Every measurement row carries a `source_ref` and a `source_location`. The location is not a page number: it names the table, the figure panel, the sheet, the column, or the registry path that the value came out of. A reader who disagrees with a number can find it at source without asking us where to look.